

FRYING PAN SHOALS BUOY 41013, USA

WMO: 997030

Lat: 33.436N

Lon: 77.743W

Elev: 10

StdP: 14.69

Time Zone: -5.00 (W05)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.2	37.9	20.2	15.2	37.7	23.5	17.8	40.9	37.3	54.8	35.9	57.1	21.7	10	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.4	84.5	79.3	84.2	79.0	83.6	78.5	81.5	83.7	80.4	83.2	79.7	82.8	13.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.5	158.6	83.4	79.6	153.4	82.9	78.5	148.0	82.4	45.0	83.9	43.9	83.1	43.1	82.8	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.9	32.0	29.4	31.9	85.3	4.3	0.6	28.8	85.7	26.2	86.1	23.8	86.5	20.6	86.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.6	54.7	56.1	59.0	65.9	71.9	78.2	80.5	81.1	78.4	71.9	64.4	60.8	(6)
(7)		DBStd	10.86	9.24	7.68	6.87	4.42	3.38	2.79	1.80	1.81	2.56	5.63	6.73	8.02	(7)
(8)		HDD50	121	63	29	12	0	0	0	0	0	0	0	3	14	(8)
(9)		HDD65	1124	332	258	205	42	3	0	0	0	0	17	93	174	(9)
(10)		CDD50	6922	208	200	290	477	679	845	944	965	851	680	433	350	(10)
(11)		CDD65	2449	11	9	18	69	217	395	479	500	401	232	74	44	(11)
(12)		CDH74	18769	3	1	14	7	386	3261	5014	5534	3367	1029	106	47	(12)
(13)		CDH80	2970	0	0	0	0	0	377	912	1314	345	22	0	0	(13)
(14)	Wind	WSAvg	15.4	17.1	17.5	17.2	15.0	13.9	13.4	13.6	12.1	14.6	15.9	17.2	16.9	(14)
(15)	Precipitation	PrecAvg	56.4	4.1	3.8	4.1	3.5	4.1	5.5	6.4	6.9	6.2	4.5	3.4	4.0	(15)
(16)		PrecMax	69.6	7.8	10.4	7.8	6.0	7.2	11.1	12.4	13.1	17.4	10.9	7.4	7.9	(16)
(17)		PrecMin	39.6	1.3	1.3	1.1	0.9	1.9	1.5	3.3	3.2	1.6	0.1	0.4	1.5	(17)
(18)		PrecStd	6.9	1.9	1.9	1.5	1.3	1.4	2.0	1.9	2.2	3.3	2.6	1.8	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.6	73.3	76.5	75.1	78.6	84.5	84.5	85.8	83.7	82.1	78.3	77.0	(19)
(20)			MCWB	69.2	68.8	72.3	71.6	73.6	78.6	79.2	80.9	77.9	77.4	73.5	73.1	(20)
(21)		2%	DB	71.6	71.3	71.7	73.9	77.9	84.1	84.2	85.0	83.0	80.5	76.6	75.6	(21)
(22)			MCWB	67.8	68.2	67.2	69.9	72.9	78.7	79.0	79.9	77.2	75.8	71.4	71.7	(22)
(23)		5%	DB	69.8	69.2	69.8	72.7	77.1	83.5	83.6	84.4	82.4	79.7	75.3	73.8	(23)
(24)			MCWB	66.3	66.2	66.0	68.6	72.2	78.9	78.6	79.2	76.5	74.9	69.6	69.5	(24)
(25)		10%	DB	67.8	67.1	68.4	71.7	76.3	82.6	83.1	84.0	81.9	78.6	73.6	71.8	(25)
(26)			MCWB	63.8	63.1	64.5	67.6	71.5	77.9	78.3	78.7	76.0	73.2	67.7	67.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	70.1	74.1	73.0	75.9	81.8	81.5	83.6	80.2	78.8	74.4	74.3	(27)
(28)			MCDWB	73.6	72.3	76.7	74.3	77.9	83.7	83.4	84.9	82.7	81.0	77.4	76.3	(28)
(29)		2%	WB	68.4	68.6	68.3	71.4	74.8	80.5	80.5	82.2	79.2	77.3	73.0	72.7	(29)
(30)			MCDWB	71.0	70.9	70.7	73.3	76.9	83.2	83.0	84.2	82.0	79.7	76.0	75.3	(30)
(31)		5%	WB	67.0	67.1	67.1	70.0	73.5	79.4	80.1	80.9	78.2	76.3	70.9	70.8	(31)
(32)			MCDWB	69.9	69.6	69.4	71.9	76.0	82.7	82.8	83.7	81.5	79.1	74.3	73.6	(32)
(33)		10%	WB	64.5	64.4	65.6	68.5	72.6	78.3	79.4	79.7	77.4	75.0	68.9	68.3	(33)
(34)			MCDWB	67.6	66.7	68.3	70.9	75.4	81.7	82.5	83.2	81.0	78.3	73.0	71.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.1	10.0	8.1	5.5	4.0	3.1	3.0	3.4	3.6	5.3	7.5	8.7	(35)
(36)			MCDBR	10.0	8.9	7.2	4.8	3.0	3.2	3.1	3.4	3.5	5.4	9.0	8.3	(36)
(37)			MCWBR	10.6	9.2	7.6	5.7	3.8	3.8	3.6	3.6	4.4	5.2	8.8	9.0	(37)
(38)		5% WB	MCDWB	9.7	8.1	6.9	5.4	3.2	2.8	3.0	3.2	3.5	5.4	9.2	7.8	(38)
(39)			MCWBR	10.3	8.9	7.1	6.5	4.2	3.8	3.9	3.8	4.1	5.3	9.4	8.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.341	0.349	0.372	0.401	0.451	0.476	0.507	0.488	0.430	0.390	0.358	0.346	(40)	
(41)		taud	2.520	2.502	2.452	2.351	2.270	2.245	2.173	2.228	2.382	2.467	2.504	2.529	(41)	
(42)		Ebn at Noon	277	286	287	282	267	259	251	254	265	268	268	268	(42)	
(43)		Edh at Noon	27	30	34	39	43	44	47	44	36	31	27	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	842	1073	1444	1846	2059	2111	1977	1768	1476	1257	943	753	(44)	
(45)		RadStd	62	109	133	138	126	136	134	118	115	120	69	52	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (23 neighbors)	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+320	+216

Nomenclature: See separate page